

# Orthographic Projection Guide

A student reference for converting a 3D object into aligned 2D views without perspective.

## Use this guide when...

- you need to communicate shape accurately without using a perspective sketch.
- you are making a multiview drawing from a part, sketch, or CAD model.
- you need to choose front, top, and right-side views that describe one object.

## Projection standard used in this guide

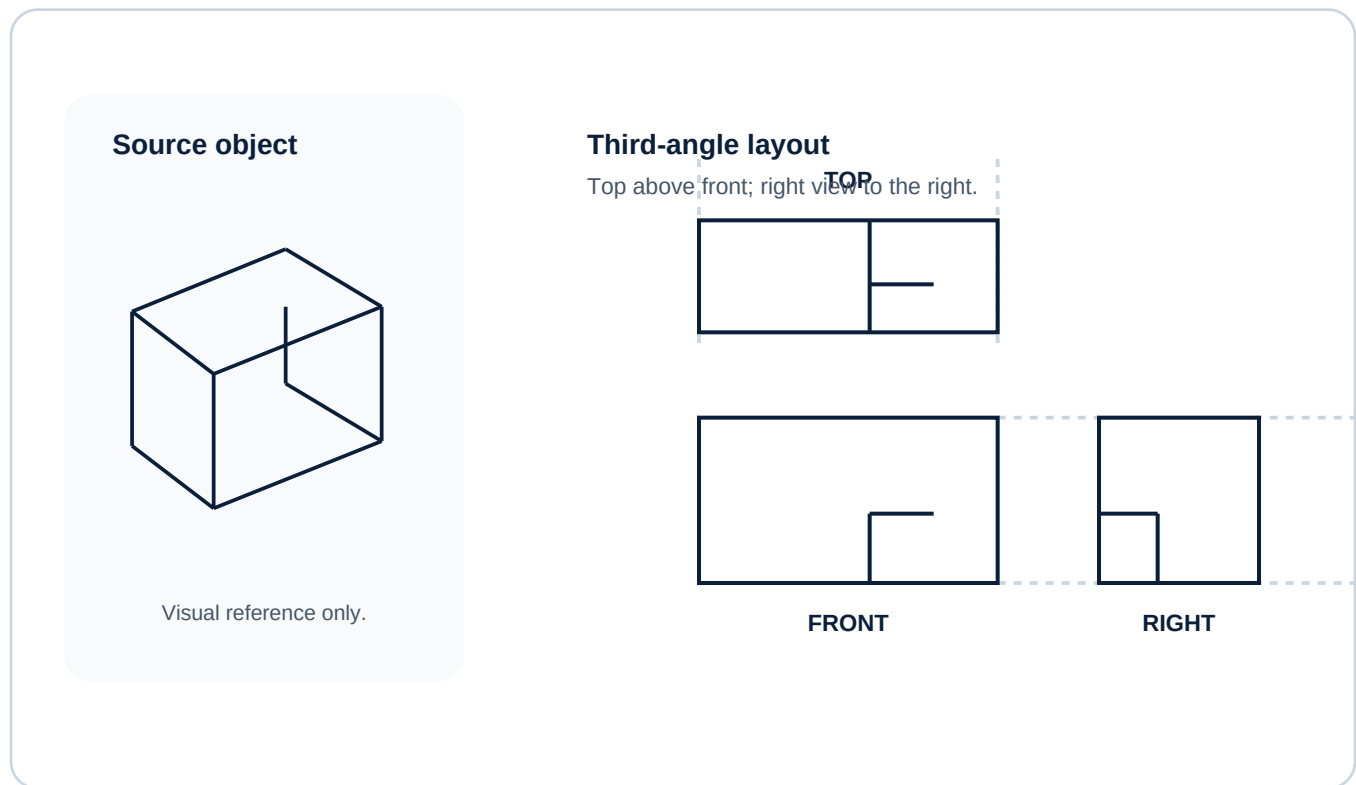
| Convention              | What it means                                                                                                  | Student check                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Third-angle projection  | The top view is placed above the front view, and the right-side view is placed to the right of the front view. | This is the layout used throughout this guide.                                                         |
| Same scale across views | All orthographic views represent the same object at the same scale.                                            | Do not stretch, shrink, or rotate one view independently.                                              |
| Aligned features        | Shared dimensions line up between related views.                                                               | Width aligns between top/front, height aligns between front/right, and depth aligns between top/right. |
| No perspective          | Orthographic views show true shape on a flat plane.                                                            | Edges do not converge like they do in a perspective sketch.                                            |

## Choosing the front view

- Choose the view that shows the most descriptive shape of the object.
- Choose the orientation that minimizes hidden lines when possible.
- Show the part in its normal use or assembly orientation when that helps the reader.
- Keep the longest or most stable dimension horizontal when it improves readability.
- After choosing the front view, do not rotate the top or right-side views independently.

# View alignment example

The three views must stay aligned because they describe the same physical object.



## Alignment rules

| Views           | Shared measurement | What to check                                                 |
|-----------------|--------------------|---------------------------------------------------------------|
| Top and front   | Width              | Left/right edges and feature locations align vertically.      |
| Front and right | Height             | Top/bottom edges and steps align horizontally.                |
| Top and right   | Depth              | Depth is transferred between the top and side views.          |
| All views       | Scale              | No view is stretched, rotated, or drawn at a different scale. |

# Hidden lines and centerlines

Use hidden lines and centerlines only when they improve the drawing.

| Line type                | Use in orthographic views                                | Accuracy note                                                                                 |
|--------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Hidden line              | Shows an important non-visible edge or feature.          | Use sparingly. A section view may be clearer for complex internal geometry.                   |
| Centerline / center mark | Identifies holes, circular features, axes, and symmetry. | A hole should usually have a center mark or centerline in the view where it appears circular. |
| Projection line          | Helps align views during layout.                         | Projection lines are construction aids; they should not look like final object lines.         |

## Common mistakes to avoid

| Mistake                                              | Better choice                                                                      |
|------------------------------------------------------|------------------------------------------------------------------------------------|
| Placing the top view beside the front view.          | Use the third-angle layout: top above front.                                       |
| Rotating the right-side view to make it fit.         | Keep the right-side view aligned with the front view.                              |
| Drawing an orthographic view with perspective edges. | Use true flat views. Parallel edges remain parallel.                               |
| Changing scale between views.                        | Use the same scale for all related views.                                          |
| Using hidden lines for every internal edge.          | Use hidden lines only when helpful; consider a section view for complex interiors. |

## Quality checklist

- The front view shows the most descriptive shape of the object.
- The top view is directly above the front view.
- The right-side view is directly to the right of the front view.
- Width, height, and depth align across related views.
- All views use the same scale and orientation.
- Hidden lines and centerlines are used only where they clarify the drawing.